

Education

University of Utah

B.S. Computer Science

Salt Lake City, UT

August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: PyTorch, React, Next.js, Node.js, AWS, SageMaker, Fargate, Docker, CI/CD, REST, Linux

Experience

University of Utah

Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT

May 2024 – Jun 2025

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

Doxy.me

Software Engineering Intern

Charleston, SC

August 2024 – May 2025

- Built and fine-tuned transformer-based models with PyTorch and Llama for real-time clinical inference, integrating WebRTC video streaming into the inference loop.
- Designed and deployed end-to-end ML inference pipelines on SageMaker and Fargate covering data ingestion, preprocessing, model serving, and post-processing; improved diagnostic accuracy using reinforcement learning.
- Implemented AI-powered document processing workflows to extract and structure clinical data from unstructured inputs, enabling downstream analytics.
- Owned CI/CD configuration across cloud infrastructure and application deployments using YAML-based pipelines, enabling reproducible environments and zero-downtime releases.

HEXstream

Software Engineering Intern

Chicago, IL

May 2022 – Aug 2022

- Engineered backend ETL pipelines integrating data from 25+ enterprise sources into Azure SQL and Azure Data Lake.
- Developed automated workflows for ingestion, cleansing, and aggregation, supporting distributed analytics systems.

Projects

University Course Planning/Review Platform: Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

Ray Tracing Engine: Created an interactive WebGL-based ray tracing engine in JavaScript, featuring realistic reflections, dynamic lighting, and customizable environment maps. Implemented shaders and user controls for rendering techniques, scene adjustments, and interactivity.